Nick Beavis

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Education

2024 – Present **Bachelor of Science**, University of Toronto, Canada
Program: *Physics Specialist*
92% average across all physics classes to date

2013 – 2016 **Doctor of Philosophy**, Loughborough University, UK
Thesis: *Numerical Modelling of Gasoline Direct Injection Engine Processes*
Supervisors: Dr. Salah Ibrahim, Prof. Weeratunge Malalasakera

2008 – 2011 **Master of Engineering**, University of Wisconsin-Madison, USA
Master of Engineering in Engine Systems

2003 – 2007 **Bachelor of Engineering (Hons)**, Loughborough University, UK
Automotive Engineering – First Class Honours

Research Interests

Theoretical early-universe cosmology, particularly inflation and particle cosmology, quantum cosmology and quantum foundations.

Research Experience

May 2026 – Sep 2026 ***Research Student***, *University of Toronto, Canada*
Research title: *Lattice simulations of SIGWs from inflation*
Supervisor: Prof. Renée Hložek
Research project using 3D lattice simulations to investigate primordial fluctuations and scalar-induced gravitational waves in α-attractor models of cosmic inflation. The project involved running and validating simulations, analysing power spectra using Python, and testing numerical predictions against linear perturbation theory.

May 2025 – Aug 2025 ***Research Student***, *University of Toronto, Canada*
Research title: *The Coalescence of Binary IMBH in Gas-Rich Star Clusters*
Supervisor: Dr. Yanlong Shi
Research project using N-body hydrodynamics simulations to investigate the coalescence of binary intermediate mass black holes in gas-rich star clusters. The project involved running large 3D simulations, post-processing and analysing large datasets using Python, and developing my understanding of galaxy dynamics.

Apr 2018 – Apr 2019 ***Research Associate***, *Loughborough University, UK*
Research position utilising 1D-cycle simulation software GT-Power to investigate the flow dynamics within a novel cylinder head concept for engineering research consultancy OakTec.

Jan 2017 – Apr 2018 ***Research Associate***, *Loughborough University, UK*
Continuation of PhD research to investigate whether the 3D-CFD predicted in-cylinder conditions at the point of sparking timing and the engine out emissions from a single-cylinder research engine could be correlated via the use of a neural network to produce a design tool for automotive manufacturer Jaguar Land Rover.

Sep 2013 – Dec 2016 ***Doctoral Student***, *Loughborough University, UK*
Full time PhD in the Low Carbon Technologies group of the Aeronautical and Automotive Engineering Department. My research involved 3D-CFD modelling and characterisation of the in-cylinder flow field and fuel injection processes using detailed computational techniques. I was also granted permission to coordinate the research project with external sponsor Jaguar Land Rover and managed additional research efforts conducted by several undergraduate students in support of the wider research goals.

Journal Articles

[1] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2018) Numerical Investigation of Combustion Regimes in a GDI Engine. *Flow, Turbulence and Combustion*. doi.org/10.1007/s10494-018-9949-8

[2] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2017) A Numerical Study of Intake Valve Jet Flapping in a Gasoline Direct Injection Engine. *International Journal of Powertrains*. hdl.handle.net/2134/25330

[3] **Beavis, N.J.**, Ibrahim, S.S. and Malalasekera, W., (2016) Impingement characteristics of an early injection GDI engine: A numerical study. *International Journal of Engine Research*. doi:10.1177/1468087416663325

[4] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2016) Numerical Simulations of a GDI Engine Flow Using LES and POD. *SAE Technical Paper 2016-01-0598*.

[5] Ahmedov, A., **Beavis, N.J.**, Pezouvanis, A.,Rogers, D., Ebrahimi, K., (2018) Design Development and Performance Evaluation of ICE Exhaust Silencer. api.semanticscholar.org/CorpusID:271602481

[6] Ahmedov, A., **Beavis, N.J.**, Kalantzis, N., Pezouvanis, A., Rogers, D., Ebrahimi, K., (2021) Method for design and evaluation of ICE exhaust silencers. *International Journal of Powertrains* doi.10.1504/IJPT.2021.114735

Thesis

[1] **Beavis, N.J.**, (2016) *Numerical Studies of Gasoline Direct Injection Engine Processes*. Loughborough University. Thesis. hdl.handle.net/2134/25230

Presentations

[1] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2017) Numerical Evaluation of Combustion Regimes in a GDI Engine. In: Tenth Mediterranean Combustion Symposium, 17-21st September 2017, Napoli, Italy.

[2] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2016) Quantification of Flow Anisotropy and Turbulence Intensity in an Early Injection GDI Engine using LES. Accepted for presentation at: Large-Eddy Simulation for Internal Combustion Engines, 30 November-1 December 2016, Rueil-Malmaison, France.

[3] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2016) A Numerical Study of Intake Valve Jet Flapping in a GDI Engine. In: 3rd Biennial International Conference on Powertrain Modelling and Control, 7-9th September 2016, Loughborough, UK.

[4] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2016) Numerical Simulations of a GDI Engine Flow Using LES and POD. 2016 SAE Congress, Detroit, USA

[5] **Beavis, N.J.**, Ibrahim, S.S., and Malalasekera, W., (2015) Characteristics of GDI Engine Flow Structures. In: Leipertz, A. (ed.) 12th International Congress, Engine Combustion Processes, Current Problems and Modern Technologies, 12-13 March 2015, Ludwigsburg, Germany. Erlangen, ESYTEC GmbH. pp. 385-396. dspace.lboro.ac.uk/2134/20016

Teaching Experience

Sep 2014 – Sep 2016 ***Teaching Assistant***, *Loughborough University, UK*
Role included teaching at undergraduate level fluid mechanics labs and marking subsequent lab report papers. Conducted this role over two academic years.

Directed Reading Groups

Sep 2025 – Present ***Gravitation: From Classical to Quantum***
A year long reading course that started by developing the mathematics and theory of general relativity, it's classical limitations and finished by exploring some of the current theories in quantum gravity.

Sep 2024 – Apr 2025 ***Fundamentals of Quantum Mechanics & Continuous Measurement***
A year long reading course that started with the fundamentals of Quantum Mechanics and finished by exploring two seminal research papers in the field of continuous measurement.

Professional Experience

Apr 2019 – May 2023 ***Contractor in System Performance Analysis***, *Cummins Ltd, Daventry, UK*

Aug 2012 – Sep 2013 ***Daventry Performance Engineering Leader***, *Cummins Ltd, Daventry, UK*

Oct 2009 – Aug 2012 ***Air-Handling Analysis-Led-Design Leader***, *Cummins Ltd, Daventry, UK*

Sep 2007 – Oct 2009 ***Combustion Performance Emissions Engineer***, *Cummins Ltd, Daventry, UK*

Sep 2005 – Sep 2006 ***Performance Development Engineering Student***, *Cummins Ltd, Daventry, UK*